Xiaosha Evelyne Li

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Researcher | Representation Learning · Audio & Multimodal Reasoning · Model Interpretability & Evaluation

EDUCATION

Georgia Institute of Technology

MSc (Thesis Track) in Computational Science and Engineering | Sept 2025 – May 2027 (Expected)
MSc in Music Technology | Sept 2025 – May 2027 (Expected)
Advised by: Prof. Pan Li (CSE), Prof. Alexander Lerch (Music Informatics Group), Ziyu Wang (Courant at NYU)
Coursework: Representation Theory, Dynamic Algebraic Algorithms, Audio Programming Engineering, Advanced Programming Techniques, Computational Data Analysis, Modeling and Simulation

Stanford University

Graduate Certificate in Electrical Engineering | May 2025 – May 2027 (Expected)
Coursework: Convex Optimization

KEY RESEARCH PROJECTS

Master's Thesis: Source-Aware Audio Understanding in Parallel Multi-Context Settings

Georgia Institute of Technology | Advised by Prof. Pan Li | In Progress

- Investigate **simultaneous understanding of multi-source, multi-context audio**, examining whether audio's concurrent, temporally aligned structure offers advantages over sequential text for parallel multi-context reasoning.

Predictive Rhythm Perception in Music Generative Models

Georgia Institute of Technology, New York University | 2026 – Present

- Identified **context-driven rhythmic prediction signals** through cross-genre token omissions, controlling codec look-ahead.
- Observed **delayed adaptation to abrupt tempo changes** in MusicGen's entropy-based tempo readouts.

Automatic Music Evaluation with Model-Intrinsic Signals — ISMIR 2026 (Accepted)

Georgia Institute of Technology, New York University, ByteDance LLC | Oct 2025 – May 2026

- Showed that a frozen music generator's own intrinsic signals — token-level loss, predictive entropy, and sparse-autoencoder (SAE) latents — predict human musicality ratings across **five benchmarks (~241 h of audio)**, without ever processing the waveform.
- Beat audio-only aesthetic verifiers by a wide margin (Pearson r=0.83 vs. 0.20 on the combined benchmark set), establishing model internals as a **stronger quality signal** than the generated audio itself.
- Traced low ratings to an **overconfident-wrong signature** in the loss–entropy structure, and showed SAE latents separate good from bad clips with no human labels — both corroborated by blind audio captioning and a professional producer.

Per-Degree Benchmark: Measuring Modality Dependence in Multimodal Benchmarks — In Submission

Georgia Institute of Technology, Smule Labs | Advised by Prof. Pan Li | May 2026 – Aug 2026

- Turned "how much does a question need its modality?" into a **per-item information degree (0–4)** — a text-only model receives increasing textual evidence (transcript → short → detailed caption), stopping at the first amount that solves the item — measured in full **across 10 audio and image benchmarks (22,160 items)**.
- Exposed design flaws previously only anecdotal: **48%** of a music benchmark solvable blind via option leakage; **77%** of a speech benchmark solvable from transcript alone.
- Validated with ten models in two groups — reasoning-enabled vs. reasoning-disabled: answers differing **47× in length** read the same degrees (blind share pinned at 17–21% for every model), and reliability climbs from ICC=0.65 for one model to **ICC=0.95 across all ten — degree is a property of the question, not the model.**
- Audited answers with a Chain-of-Thought (CoT) observer that investigates a 25.8M-token reasoning corpus by **retrieving only 0.05%**, attributing each unsolved item to missing perception vs. missing knowledge.

Rethinking Likelihood-Based Evaluation in Music Large Language Models — ICASSP 2026 (Published)

Georgia Institute of Technology, New York University, ByteDance LLC | May 2025 – Sept 2025

- Studied autoregressive music LLMs, addressing the gap between statistical likelihood and musical perception.
- Built a noise-injection framework to test how models recover structure under token corruption.
- Identified a **Context Amnesia Effect** reflecting breakdowns in temporal signal structure and audio feature representation under local perturbations.
- Proposed a **label-free, profile-based evaluation** aligning model reliability with perceptual structure.

Spectral Probing of Frozen Audio Encoders

Georgia Institute of Technology | Advised by Prof. Alexander Lerch | Jan 2026 – May 2026

- Probed **four frozen audio encoders (CLAP, AudioMAE, BEATs, PaSST)** on **four MIR tasks** spanning clip-level to per-frame timescales, under two pre-registered hypotheses.

- Discovered **spectral concentration**: task-relevant information concentrates in the lowest temporal-frequency band regardless of task timescale — overturning the NLP-derived expectation that band should track task speed.
- **Ruled out artifacts** with four null perturbations (random-init encoder, label shuffle, DC removal, time permutation), isolating architectural and temporal-ordering contributions.

System-Level Diversity Evaluation of Music LLMs

Georgia Institute of Technology | Advised by Prof. Alexander Lerch | May 2025 – Dec 2025

- Built a scalable, system-level diversity evaluation pipeline for music LLMs, integrating source separation, representation extraction, caption-based prompting, regeneration, and distributional analysis.
- Deployed Qwen3-Omni via vLLM on 4×H100/H200 GPUs for consistent, high-throughput evaluation across multiple generative model systems.

RESEARCH & INDUSTRY EXPERIENCE

Research Scientist Intern

Smule Labs | San Francisco, CA | May 2026 – Present

- Built the end-to-end synthetic data generation pipeline behind a **6T-token corpus** spanning three audio domains — **speech, SFX, and music** — plus QA-pair generation, covering prompt construction, generation, filtering, and category assignment.
- Stood up and benchmarked inference architectures **across seven GPU platforms (4090-S/SW, 5090-S/SW, A100, H100, H200)**, converging on an optimized **8×RTX 5090 cluster serving Qwen3.6-35B-A3B-AWQ at ~2,000 tokens/s** for generation and evaluation workloads.
- Designed and populated **benchmark databases of 200k audio and 100k image items** with hierarchical, cross-layer indexing, powering QA-pair generation and linking synthetic pairs to their counterparts in real benchmarks.
- Built the lab's internal benchmarking and visualization platform for evaluating in-house algorithms, used directly by non-engineering stakeholders.
- Led the **Per-Degree Benchmark** research project on this infrastructure (see Key Research Projects).

CONFERENCE PUBLICATIONS

Submitting: Li, X., Zang, Y., Yang, Q., & Li, P. (2026). How Much of the Modality Does a Question Need? A Per-Item, Reliability-Certified Measurement of Multimodal Benchmarks.

ISMIR 2026 (Accepted): Li, X., Liu, C., & Wang, Z. (2026). Do Music Generative Models Understand Musicality? Automatic Music Evaluation with Model-Intrinsic Signals. *Proceedings of the 27th International Society for Music Information Retrieval Conference (ISMIR 2026)*, Abu Dhabi, UAE.

ICASSP 2026 (Published): Li, X., Liu, C., & Wang, Z. (2026). When Noise Lowers the Loss: Rethinking Likelihood-Based Evaluation in Music Large Language Models. *Proceedings of the 51st IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP 2026)*, Barcelona, Spain.

ICMC 2025 (Poster): Li, X., & Leonard, N. (2025). Algorithm Based GEN Video to Partial GEN Audio Synthesis System. *Proceedings of the 50th International Computer Music Conference (ICMC 2025)*, Boston, USA.

Technical Report (Working Paper): Li, X., & Lerch, A. (2026). Spectral Probing of Frozen Audio Encoders: An Exploratory Study of Low-Frequency Concentration.

AMIF 2025 Panel — Moderator & Organizer: "AI + Music: Empowering, Not Overpowering," a cross-sector panel with leaders from Meta, Georgia Tech, NYU, CMU, and Fish Audio.

Peer Review: Reviewer, ISMIR 2026 and ICASSP 2027; Paper Review Committee, ICMC 2025 & 2026 — music generation, audio synthesis, and evaluation.

CORE SKILLS

Programming: Python, C++, Java, MATLAB, JavaScript, Max/MSP

ML & Systems: PyTorch, vLLM, Transformers, Generative Models, Sparse Autoencoders, Multimodal Embedding Learning, Representation Learning, Multi-GPU Deployment & Inference Optimization

Audio & Signal Processing: DSP, Fourier & DCT Spectral Analysis, Source Separation, Audio Feature Representation, Vocal Synthesis, Embedded Systems & Microcontrollers

Mathematics: Linear Algebra, Convex Optimization, Probability & Statistics, Representation Theory

HONORS & CERTIFICATIONS

Berklee Composition Department Scholarship – Amy Beach Award | March 2022
Honorable Mention, MTNA Competition (3rd place, Massachusetts) | September 2021
First Place, High-Level Art Troupe Exams (Hunan University, Beihang University) | 2015
Ambassador for Cultural Exchanges between China and America | 2012
Ambassador for Cultural Exchanges between China and Australia | 2012